

Data-607-Project1--LLM Transcript

Bikash Bhowmik 27 Sep 2026

LLM Transcript

Tool Used: ChatGPT (OpenAI, GPT-5.2)

Purpose: Designing and validating relational structure from a semi-structured chess tournament cross-table

Course: DATA 607 – Project 1

Overview of LLM Usage

I used ChatGPT to assist with:

1. Structuring the parsing strategy for multi-line player records.
2. Designing the opponent extraction logic using regular expressions.

I worked iteratively, running and testing all code locally before finalizing.

[1] Conversation with ChatGPT (Parsing Strategy Design)

How should I parse a semi-structured tournament cross-table where each player spans two lines?

Assistant (ChatGPT):

The correct approach is to: - Identify line 1 records using a pattern like `^\s*\d+\s*\|` - Identify line 2 records using state abbreviations `^[A-Z]{2}\|` - Reconstruct player-level records by binding corresponding lines -

Extract pair number, player name, total points, and raw round

data - Extract pre-rating from the second line using a pattern like `R:\s*([0-9]+)`

This approach ensures that parsing is rule-based rather than position-based, making the workflow more robust to spacing variations.

[2] Conversation with ChatGPT (Opponent Extraction Logic)

How should I extract opponent pair numbers while excluding byes and nongames?

Assistant (ChatGPT):

Use a regular expression that captures only valid game patterns: `(W|L|D)\s*(\d+)`

This ensures: - Only games with a result indicator are included. - Tokens like H, B, U, or X are automatically excluded. - The denominator equals the number of actual games played.

Suggested implementation:

```
extract_opponents <- function(x) {  
  opp <- str_match_all(x, "(W|L|D)\\s*(\\d+)")[[1]]  
  if (nrow(opp) == 0) return(integer(0))  
  as.integer(opp[,3])  
}
```

[3] Conversation with ChatGPT (Validation and QA Design)

How should I validate correctness before trusting results?

Assistant (ChatGPT):

Implement two levels of validation:

Manual test cases:

- One player who played all rounds.
- One player with byes or missing rounds.
- Hand-calculate opponent rating average.

Compare with computed output.

```
stopifnot(nrow(players) == 64)  
stopifnot(all(all_opps >= 1 & all_opps <= 64))  
stopifnot(!any(is.na(unlist(players$opponent_pre_ratings))))
```

This enforces structural integrity and prevents silent logical errors.